

PORTFOLIO



DATA SCIENCE & ENGINEERING

Data Analytics | Data Engineering
AI & Automation | MLOps

Presented By: Farrel Julio Akbar

HELLO, I'M FARREL JULIO AKBAR

DATA ENGINEERING | DATA ANALYTICS | DATA SCIENCE

AI, AUTOMATION & MLOPS

Data professional with hands-on experience in data engineering, data analytics, data science, AI, and workflow automation across telecommunications and central banking. Experienced in developing near real-time data pipelines, RAG-based applications, machine learning workflows, monitoring dashboards, and data-driven decision-support systems.

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github.com/Julio-analyst



EDUCATION



Institut Teknologi Sumatera

Bachelor of Data Science | 2022–2026
GPA: **3.00/4.00**

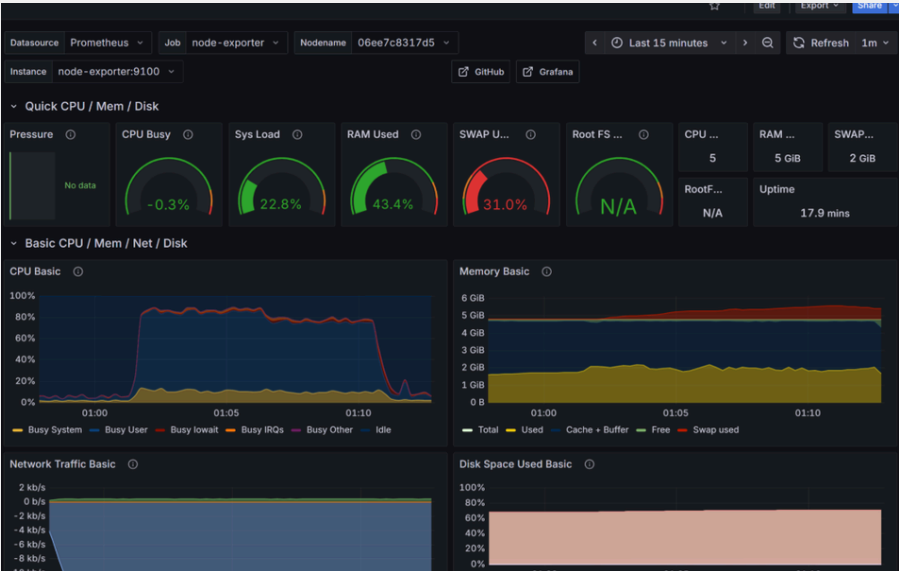
Final Project

“Performance Evaluation of PeerDB and Debezium in Docker-Based Change Data Capture Simulations”

Benchmarked near real-time PostgreSQL-to-ClickHouse CDC pipelines across workloads of 5 to 80 million rows. Evaluated latency, throughput, CPU utilization, memory utilization, scalability, and reliability using Docker and DigitalOcean.

Relevant Coursework

- Big Data Analytics
- Deep Learning
- Business Intelligence
- Data Engineering
- Database Systems
- Statistics
- Machine Learning
- Cloud Computing
- Data Visualization



Data Science & Artificial Intelligence

Developing machine learning, deep learning, NLP, RAG, and analytical modeling solutions, including feature engineering, model evaluation, error analysis, and AI workflow automation.

Data Engineering & Big Data

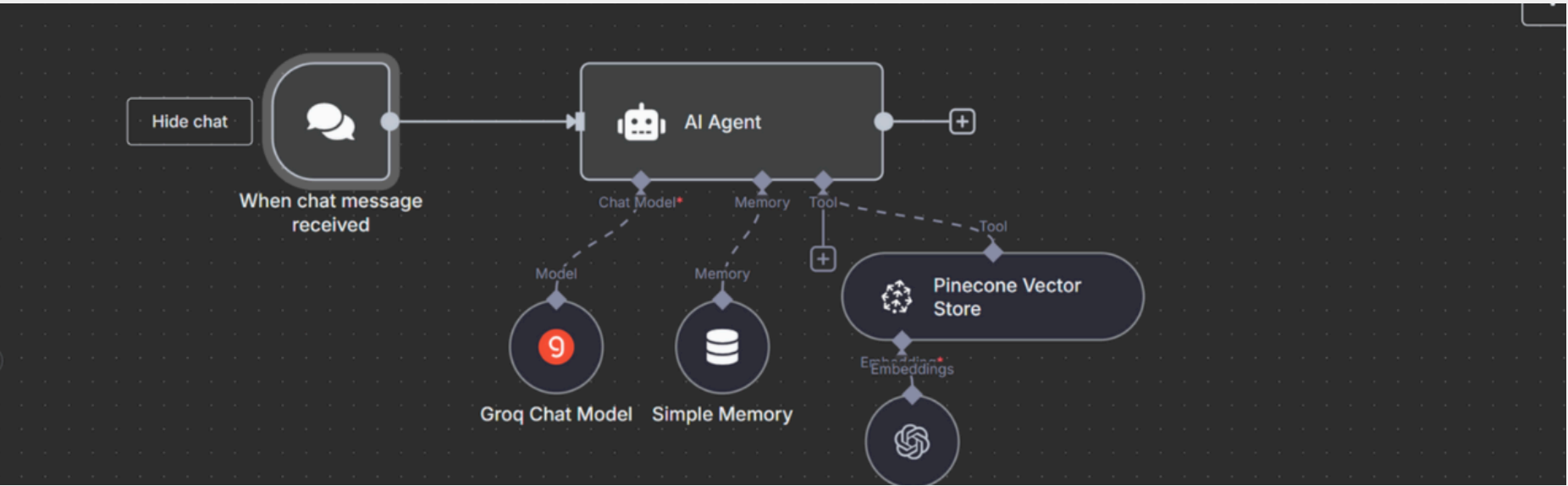
Developing ETL/ELT workflows, near real-time Change Data Capture pipelines, streaming architectures, and analytical data layers using PostgreSQL, ClickHouse, BigQuery, PeerDB, Debezium, Kafka, Flink, Airflow, and Docker.

Data Analytics & Business Intelligence

Performing data cleaning, exploratory analysis, statistical analysis, time-series analysis, forecasting, and dashboard development using Python, R, SQL, Power BI, Tableau, and Looker Studio.

MLOps & Monitoring

Implementing experiment tracking, data and model versioning, CI/CD, containerized deployment, and service monitoring using MLflow, DVC, GitHub Actions, Docker, Flask, Prometheus, and Grafana.



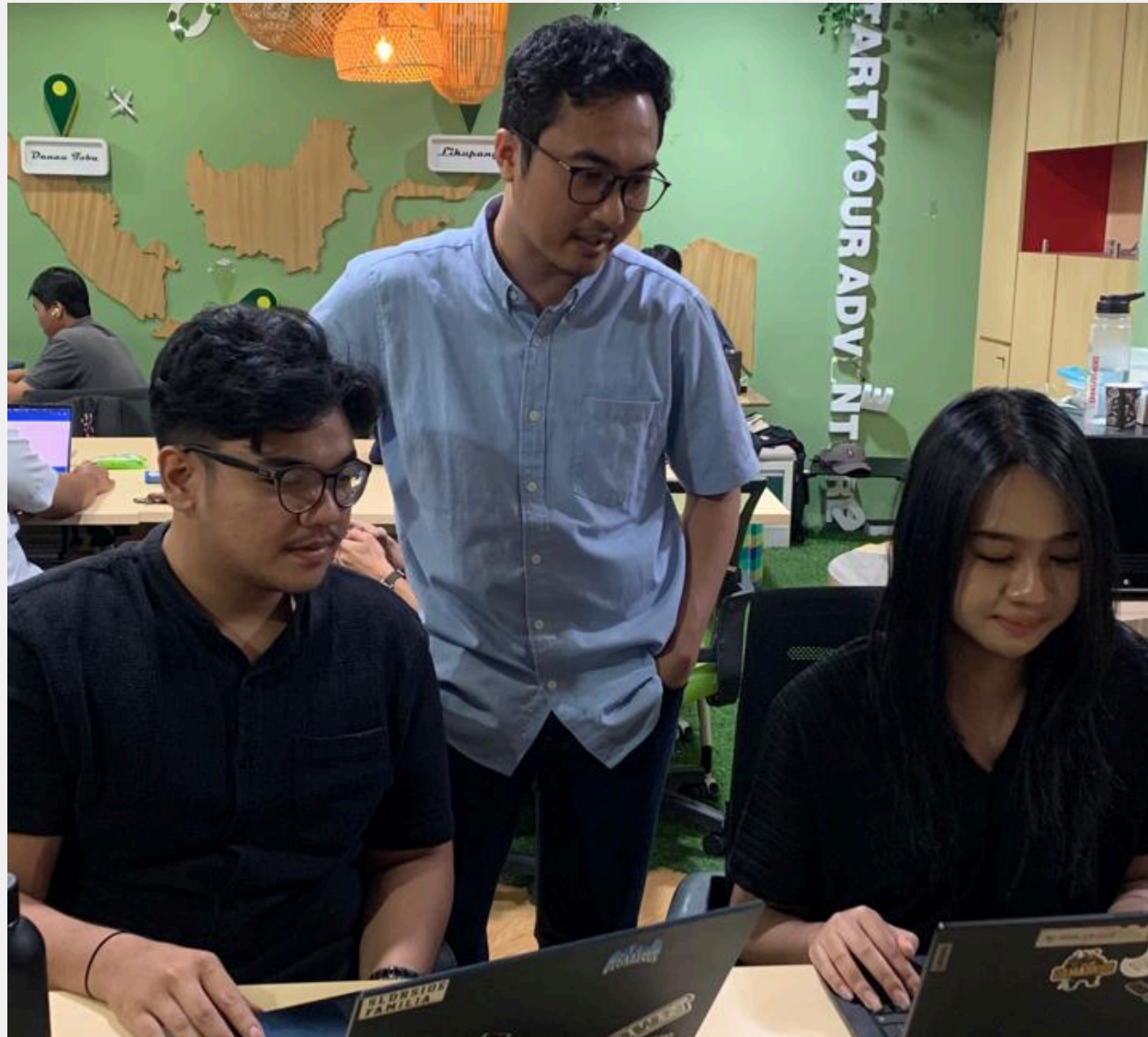
TECHNICAL EXPERTISE

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WORK EXPERIENCE



IT Specialist Intern at Rajawali Academy 2024

- Developed an automated assessment and scoring workflow using Google Sheets and Autocrat.
- Supported digital infrastructure, website operations, and the management of more than 15,000 product records.
- Applied data and digital tools to support operational efficiency and online sales activities.

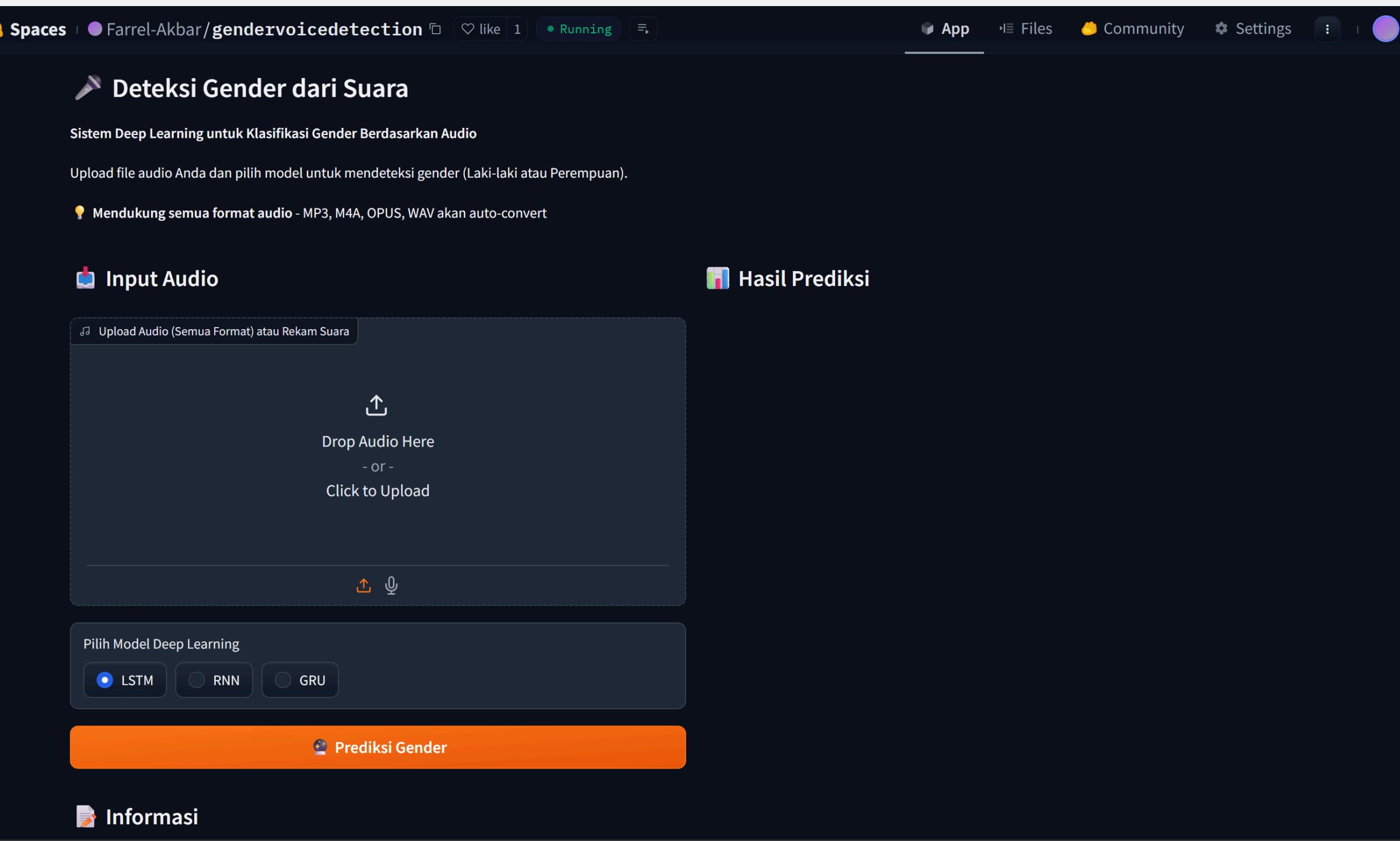


Data Scientist Intern at Telkom Indonesia 2025

- Developed a RAG-based chatbot and automated knowledge-retrieval workflow using n8n, Supabase, OpenAI, Groq, BigQuery, and Streamlit.
- Built near real-time PostgreSQL-to-ClickHouse CDC pipelines using PeerDB, Debezium, Kafka, and Apache Flink.
- Monitored pipeline latency, throughput, database performance, and system reliability using Grafana and Percona.

Data Science Intern at CENTRAL BANK INDONESIA May 2026 - Present

- Analyzed inflation, food prices, GRDP, sectoral developments, survey results, and regional economic indicators across 15 regencies and cities in Lampung.
- Developed an automated Food Price Monitoring Dashboard and Early Warning System for nine strategic commodities using PIHPS data, statistical indicators, Python, Power BI, and Looker Studio.
- Conducted time-series, business-cycle, and forecasting analysis to support regional economic monitoring and data-driven assessment.



GENDER VOICE DETECTION END-TO-END MLOPS PLATFORM

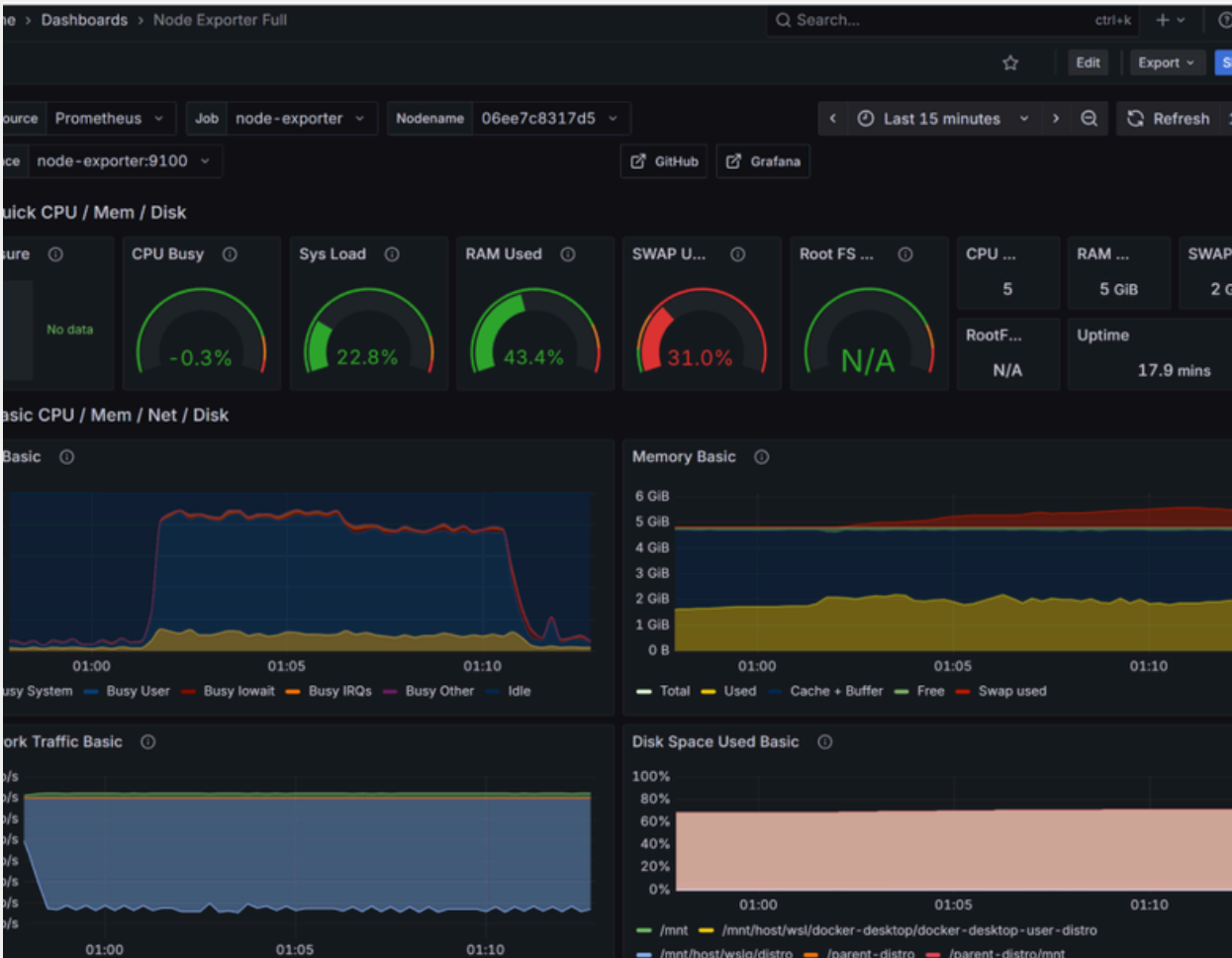
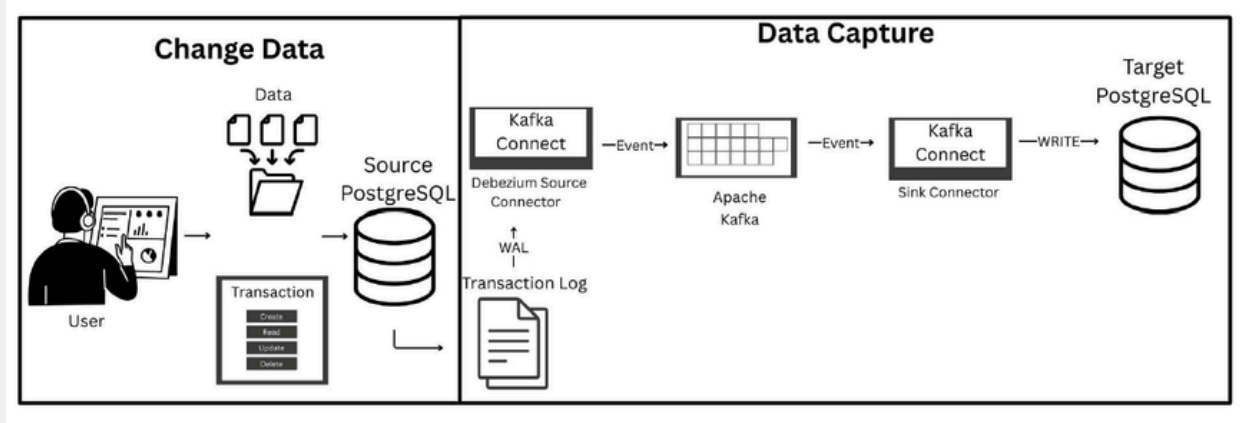
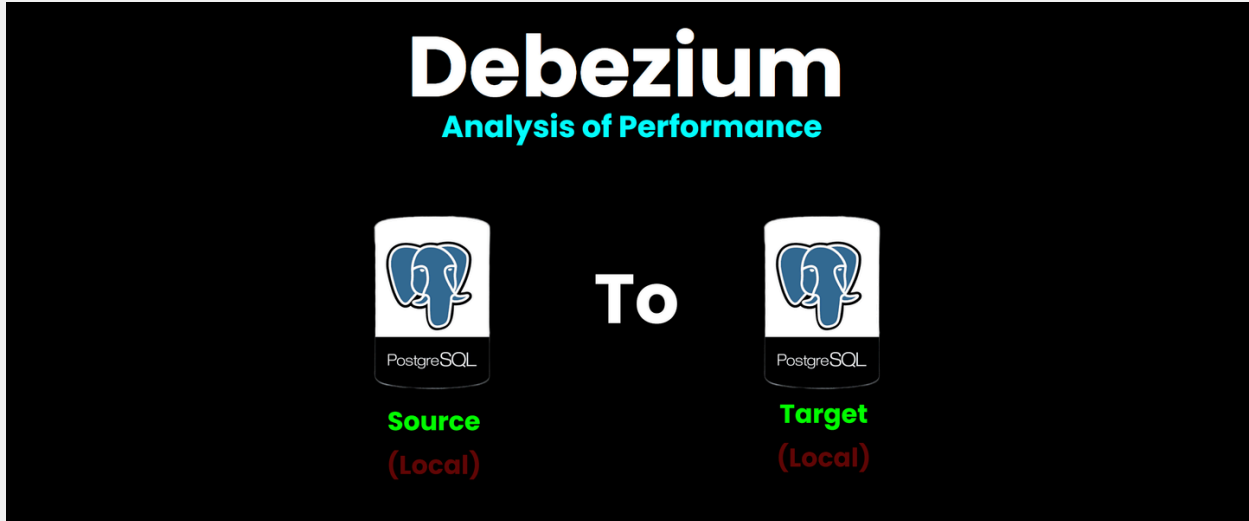
Deep Learning-based audio classification with production MLOps pipeline

End-to-end MLOps platform for gender classification from voice recordings.

The system processes audio into MFCC features, supports multiple deep learning models, and enables real-time inference with automated training, tracking, and deployment.

Tech Stack

- Deep Learning: TensorFlow, Keras (RNN, LSTM, GRU)
- Audio Processing: Librosa, MFCC
- MLOps: MLflow, DVC, GitHub Actions
- Backend & UI: FastAPI, Gradio
- Deployment: Docker, Hugging Face Spaces

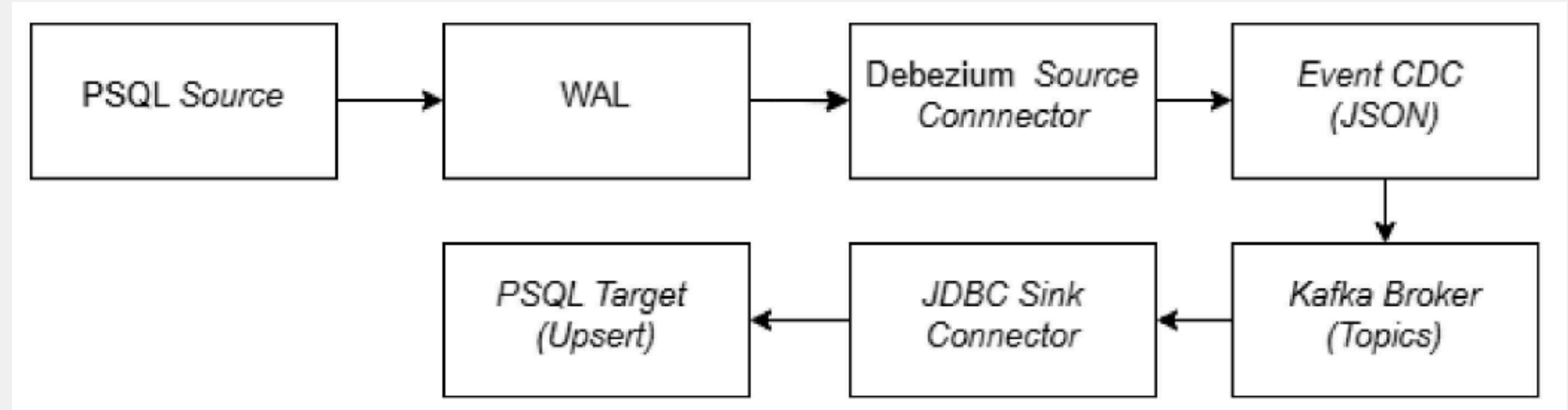


DEBEZIUM CDC MIRRORING WITH GRAFANA MONITORING:

Real-time PostgreSQL replication and observability pipeline

Designed and implemented a Change Data Capture (CDC) pipeline using Debezium to mirror PostgreSQL data changes in real time.

The system captures insert, update, and delete events from the source database, streams them to the target database, and provides real-time monitoring and performance insights through Grafana dashboards.



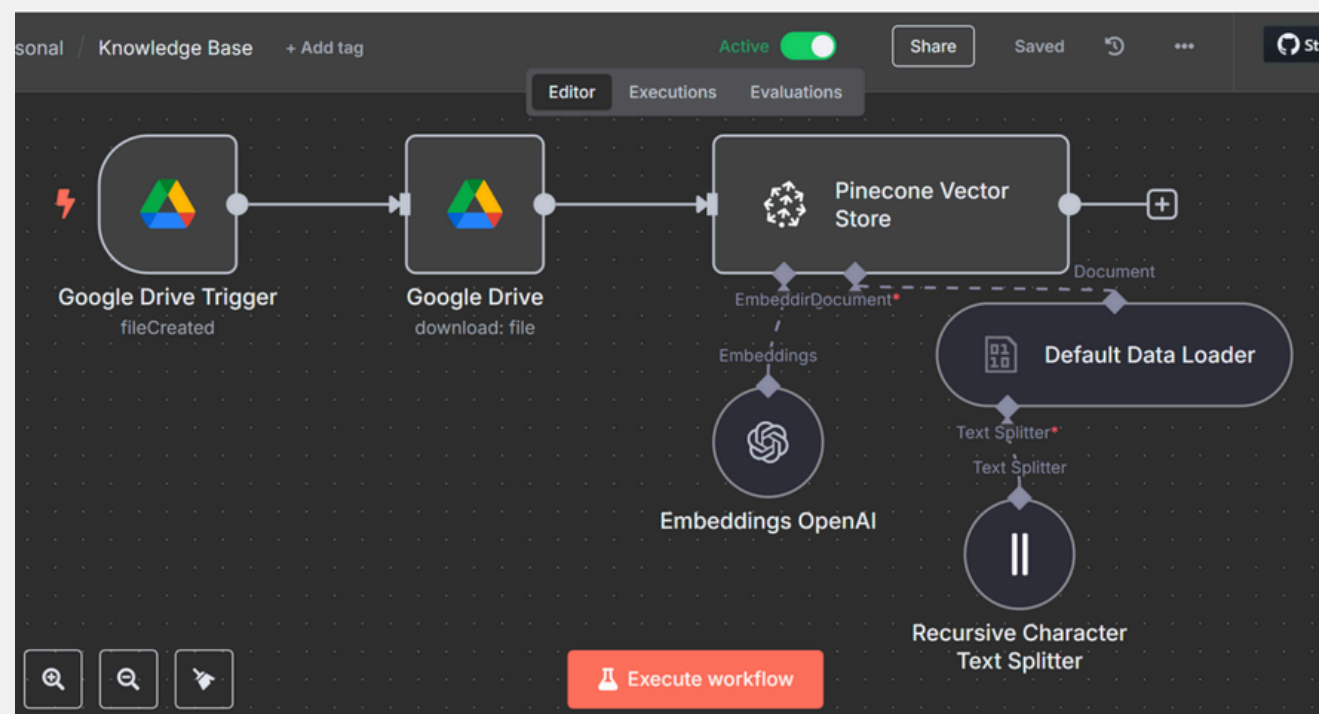
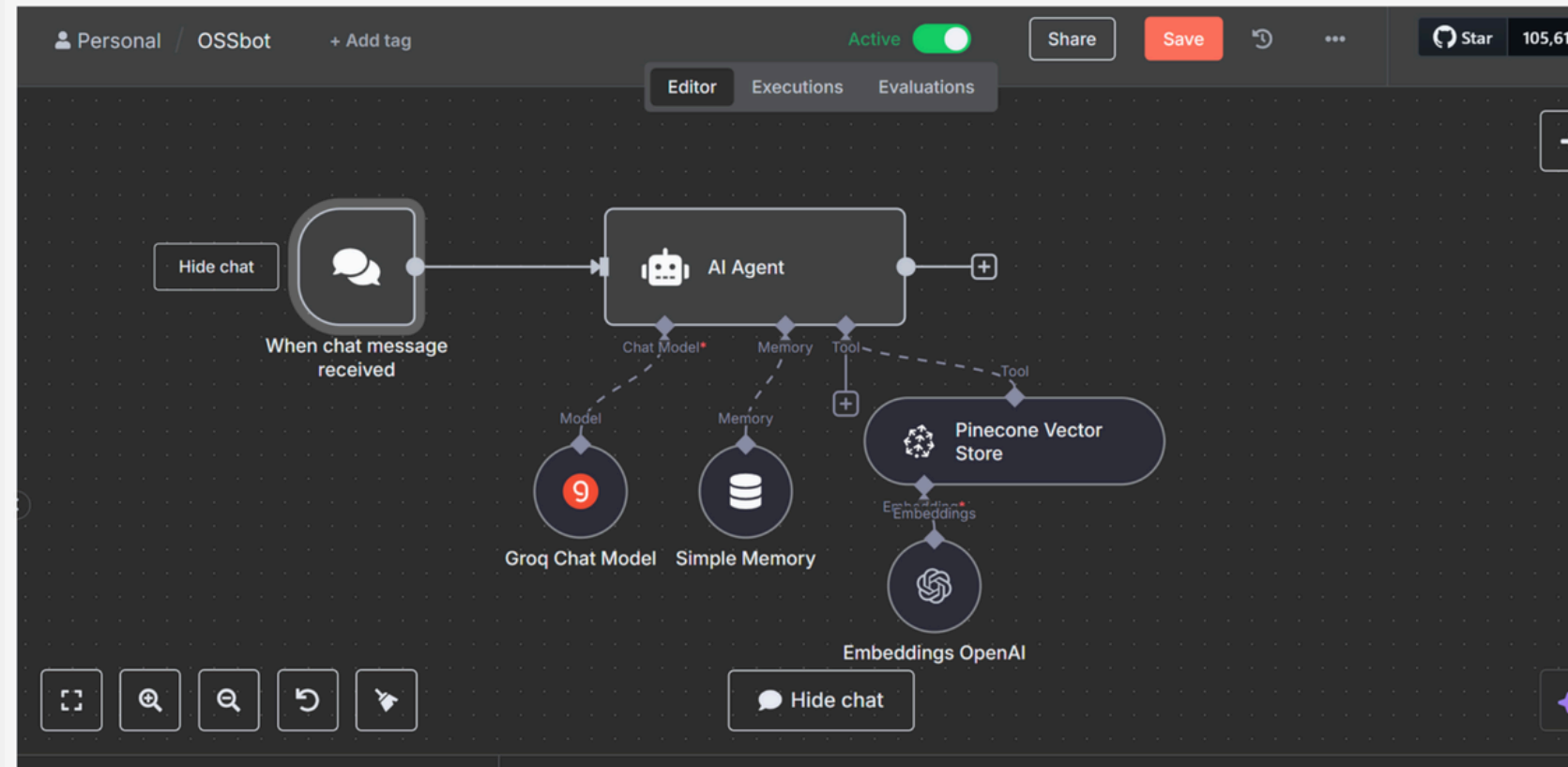
Tech Stack

- CDC Engine: Debezium
- Databases: PostgreSQL (Source & Target)
- Streaming & Connectors: Kafka / Kafka Connect
- Monitoring: Grafana
- Metrics & Observability: PostgreSQL Metrics / Custom Queries
- Environment: Docker

GitHub: github.com/Julio-analyst/debezium-cdc-postgres

RAGCHATBOT AUTOMATION USING N8N WORKFLOW

End-to-end Retrieval-Augmented Generation chatbot
with automated data ingestion



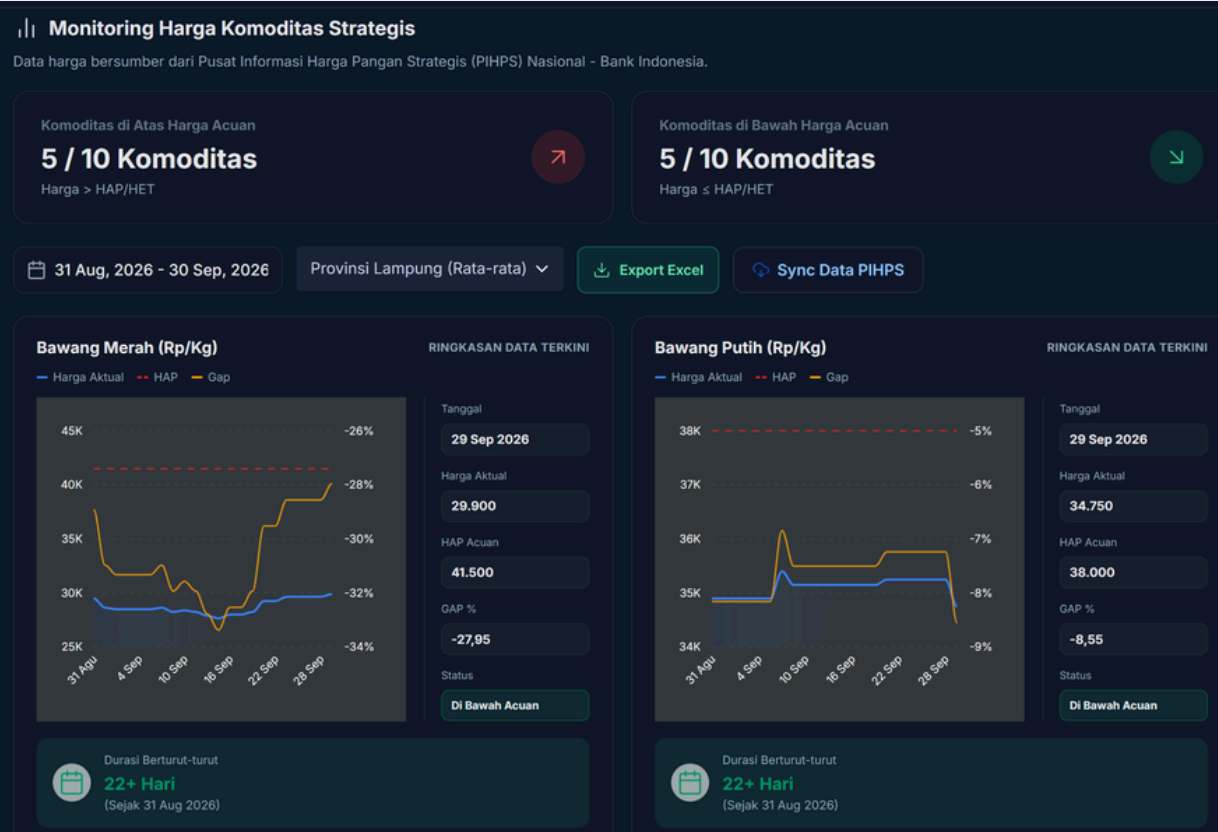
Designed and implemented an automated RAG-based chatbot using n8n to handle document ingestion, vectorization, and conversational AI workflows.

The system integrates cloud storage, vector databases, and large language models to deliver contextual, accurate responses while supporting scalable automation and monitoring.

Tech Stack

- LLM & RAG: OpenAI, Groq
- Vector Database: Pinecone
- Workflow Automation: n8n
- Data Ingestion: Google Drive, Google Sheets
- Embeddings & Indexing: OpenAI Embeddings
- Storage & Analytics: BigQuery, Supabase





Monitoring Pangan

The Food Price Monitoring panel provides an interactive view of price developments across commodities and periods. It enables users to monitor current price levels, daily and monthly movements, historical trends, and unusual changes through filters, charts, and summary indicators.

Data Source: PIHPS

Data Processing: Python, Pandas, NumPy

Statistical Methods: CQGR, CAGR, Rolling Volatility, PCA, IPA

Visualization: Power BI, Looker Studio

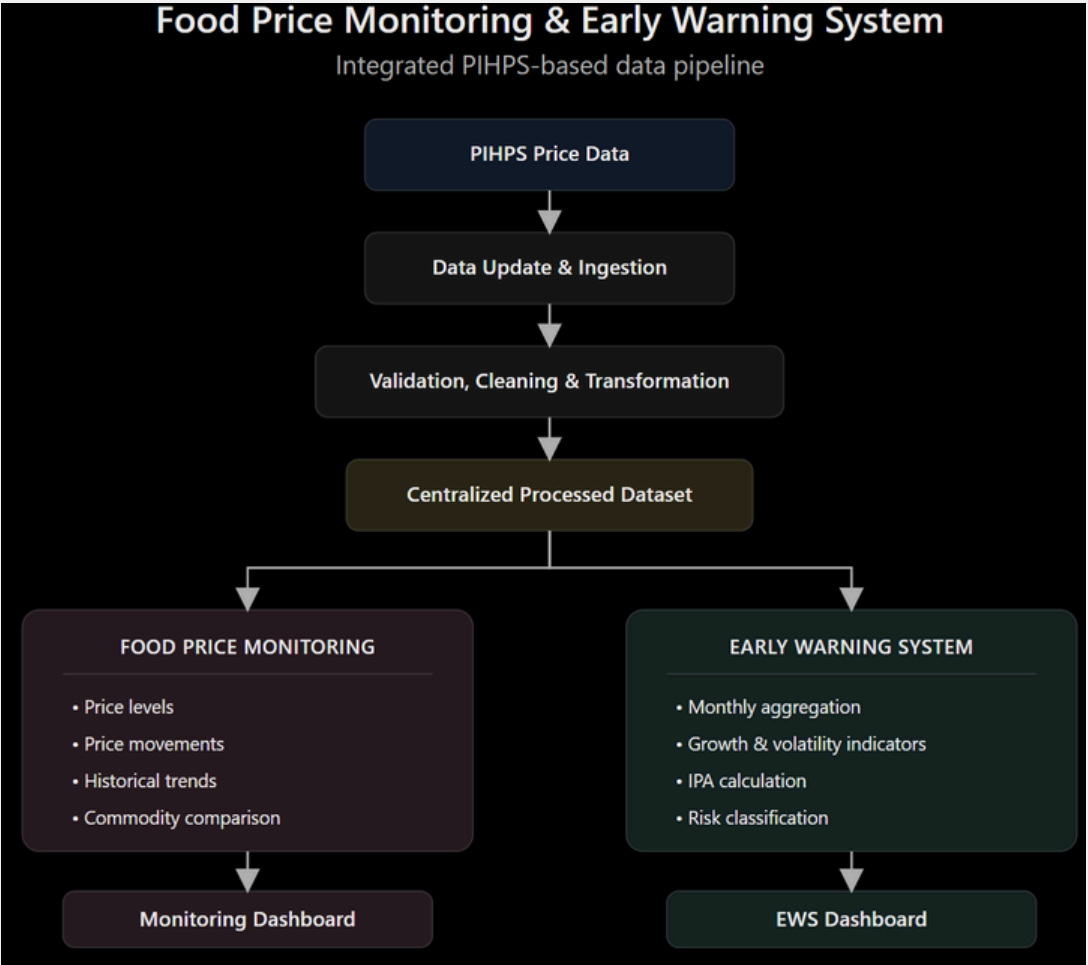
- **Outputs:** Food Price Monitoring Panel and EWS Risk Panel

FOOD PRICE MONITORING & EARLY WARNING SYSTEM

Integrated PIHPS-based price monitoring and decision-support system.

Project overview

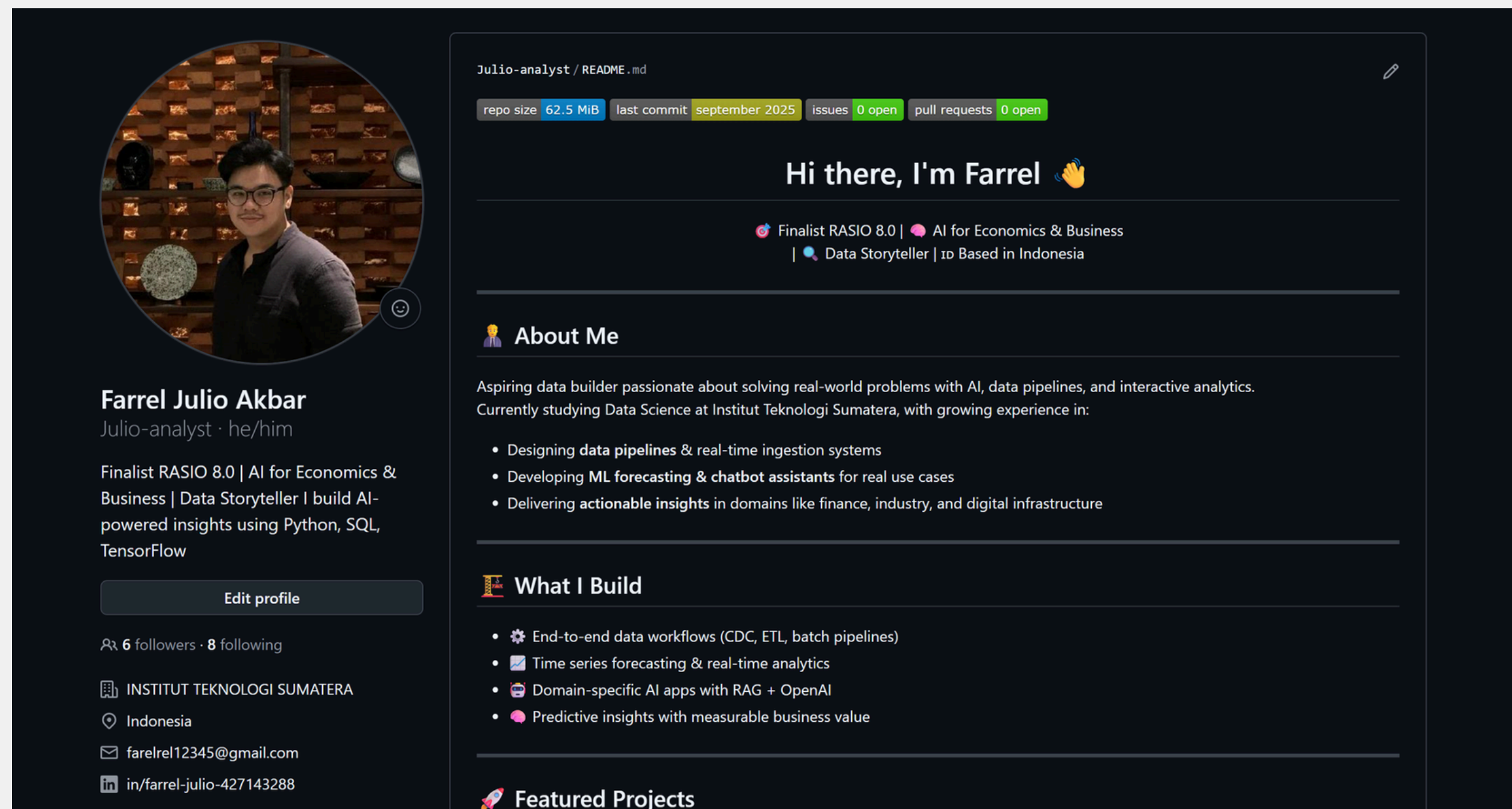
Developed an integrated food price monitoring and Early Warning System using PIHPS data for nine strategic commodities. The system processes updated price data through validation, cleaning, transformation, and structured storage before distributing it into two analytical panels. The Food Price Monitoring panel presents price levels, movements, historical trends, and commodity comparisons, while the Early Warning System applies statistical growth and volatility indicators to calculate the Indicator of Price Anomalies and classify emerging price risks.



Early Warning System

The Early Warning System aggregates daily PIHPS data into monthly observations and calculates CQGR, CAGR, rolling volatility, volatility-adjusted growth, quarterly IPA, and annual IPA. PCA-derived weights are applied to generate a composite IPA score, which classifies commodity conditions into Normal, Alert, and Warning levels for early risk monitoring.

PROJECT REPOSITORY



Explore all my projects and technical work on GitHub and LinkedIn.

<https://github.com/Julio-analyst>

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THANK

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You

Thank you for reviewing my portfolio.
I would be happy to discuss my work and experiences further.

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